

EduPath: A Personalized Study Guidance System for Engineering Students

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Abstract—Engineering students frequently struggle to manage vast syllabi under exam pressure, compounded by the absence of personalized academic guidance. This paper presents EduPath, a web-based personalized study guidance system that assists students by generating adaptive study plans tailored to individual learning needs. EduPath assesses student performance through module-level knowledge quizzes, classifies each student into one of three levels—Basic, Intermediate, or Advanced—and generates a personalized study timetable with level-appropriate learning resources including PDF notes and YouTube video tutorials. The system further incorporates a real-time student chat room for peer interaction and collaborative doubt clarification. Built using React.js for the frontend and Python Flask for the backend, with SQLite for data persistence, EduPath provides a scalable, accessible, and effective platform to help engineering students organize their studies and prepare efficiently for examinations.

Index Terms—Personalized Learning, Study Planning, Adaptive E-Learning, Knowledge Assessment, Timetable Generation, Flask, React.js, Engineering Education

I. INTRODUCTION

Engineering students face significant challenges in managing large and complex syllabi within constrained time frames. The diversity of subjects, variation in individual aptitude across modules, and the pressure of approaching examination schedules make effective self-directed study planning a difficult task. Without personalized guidance, students frequently allocate study time inefficiently—over-preparing in areas of strength while neglecting weaker topics—ultimately compromising their overall academic performance.

Existing e-learning platforms often deliver one-size-fits-all content that does not account for individual differences in prior knowledge or learning pace. There is a clear need for systems capable of dynamically assessing a student's current knowledge level and generating adaptive study plans that prioritize topics requiring the most attention, while recommending resources calibrated to the student's proficiency.

EduPath is a personalized study guidance system developed to address these gaps. The platform allows students to attempt module-level knowledge assessment quizzes across their enrolled subjects. Based on quiz performance, each student's knowledge level per module is classified as Basic, Intermediate, or Advanced. This classification drives two core

outputs: a personalized study timetable that allocates daily study hours proportional to the identified knowledge gaps, and a curated set of learning resources—PDF study materials and YouTube video tutorials—matched to the student's level in each module.

Additionally, EduPath provides a live chat room that enables students to communicate in real time, fostering peer collaboration and collective doubt resolution. An administrative interface allows teachers or administrators to manage subjects, modules, quiz questions, and study materials, ensuring that the platform's content remains current and relevant.

The system is implemented with a React.js frontend, a Python Flask backend, and an SQLite database managed via SQLAlchemy ORM. EduPath demonstrates how lightweight, accessible web technologies can deliver meaningful personalization in academic study support.

II. LITERATURE REVIEW

Kyewski and Krämer [1] proposed a Dynamic Learning Plan Generator for smart learning environments that uses algorithmic scheduling to create personalized learning plans based on learner progress and upcoming deadlines. Their system demonstrated improvements in course completion rates and learner motivation. However, the approach requires continuous integration with real-time learning management data, which increases system complexity and limits deployment in resource-constrained environments.

Chang et al. [2] developed an assessment-centered e-learning system using personalized dynamic assessments and adaptive learning materials delivered through a web-based interface. The system reduced student misconceptions and improved learning achievement scores in controlled evaluations. A key limitation identified was the dependency on carefully designed assessment items and accurate automated evaluation mechanisms to ensure reliable level classification.

Oikarinen and Reed [3] examined Internet instant messaging and public chat protocols based on client-server architecture, identifying common features such as chat rooms, user authentication, and real-time messaging. While their work predates modern web frameworks, it established foundational principles

for secure and scalable real-time communication—principles reflected in EduPath’s live chat module.

Shute and Towle [4] explored adaptive e-learning using Aptitude-Treatment Interaction (ATI) theory to personalize learning content based on individual learner characteristics and measured performance. The work demonstrated that content personalization leads to significant improvements in learning effectiveness and student engagement. A noted challenge was the complexity and cost associated with constructing accurate learner models, a concern that EduPath addresses through a lightweight quiz-based classification approach.

III. METHODOLOGY

EduPath is structured around a client-server architecture comprising a React.js frontend, a Flask-based backend, and an SQLite database. The system is organized into six functional components that together deliver an end-to-end personalized study guidance experience.

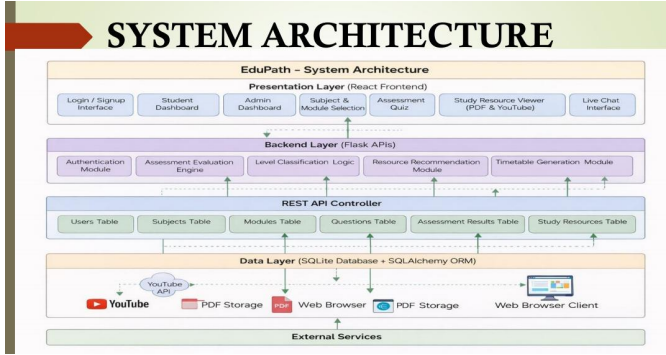


Fig. 1. System Architecture of the EduPath Platform.

A. Requirement Analysis

System requirements were identified for two primary user roles: students and administrators. Core features defined during this phase include the quiz-based knowledge assessment system, automated timetable generation, level-based learning resource recommendation, and a real-time peer chat facility. The admin role encompasses subject management, module creation, question authoring, and student performance monitoring.

B. System Design

A layered client-server architecture was designed with clear separation between the presentation layer (React frontend), the business logic layer (Flask APIs), and the data layer (SQLite via SQLAlchemy ORM). The database schema was designed to store user profiles, subjects, modules, quiz questions, assessment results, study resources, and chat messages. User interface flows were designed for student login, quiz-taking, timetable viewing, resource access, and chat interaction, as well as corresponding admin screens for content management.

C. System Development

The frontend was developed using React.js, HTML, and CSS, providing a responsive and intuitive interface for both

students and administrators. The backend was implemented in Python using the Flask microframework, exposing RESTful API endpoints consumed by the frontend. Core backend modules include an Authentication Module for secure user registration and login, an Assessment Evaluation Engine for automatic quiz scoring, a Level Classification Logic module that determines knowledge level per module based on score thresholds, a Resource Recommendation Module that retrieves appropriate PDF and YouTube resources based on the classified level, and a Timetable Generation Module that computes personalized daily study schedules.

D. Database Implementation

The SQLite database stores all persistent application data across six primary tables: Users, Subjects, Modules, Questions, Assessment Results, and Study Resources. SQLAlchemy ORM is used to abstract database interactions, ensuring clean separation between application logic and data access. Chat messages are also persisted to support message history retrieval.

E. Timetable Generation Logic

The personalized timetable is generated by analyzing a student’s knowledge level classification across all modules of each enrolled subject. Students classified at the Basic level are allocated more daily study hours for the corresponding module, reflecting a greater need for foundational coverage. Intermediate-level students receive a moderate allocation, while Advanced-level students are assigned reduced time, allowing them to focus effort on weaker areas. The timetable also factors in the number of days remaining until each subject’s examination date, enabling urgency-aware scheduling.

F. Real-Time Chat Room

A live student chat room was implemented to support peer-to-peer communication and collaborative doubt resolution. The chat module enables students to send and receive messages in real time, facilitating an informal yet structured channel for academic discussion and study group interaction.

G. Testing and Deployment

The integrated system was tested across all functional modules, including user authentication, quiz evaluation accuracy, knowledge level classification correctness, timetable generation logic, resource retrieval, and chat functionality. The system was deployed on a local development server for user access and evaluation during the project phase.

IV. FUNCTIONAL REQUIREMENTS

A. User and Admin Authentication

The system provides separate login interfaces for administrators and students. Users can register with their credentials, and all authentication is handled securely by the backend authentication module.

B. Subject and Module Management

Administrators can add subjects with associated examination dates, and create or delete modules within each subject. This content management capability ensures that the platform's academic structure remains aligned with the current curriculum.

C. Quiz and Assessment System

Administrators can author multiple-choice quiz questions with configurable difficulty levels (Basic, Intermediate, Advanced) for each module. The system conducts quizzes, evaluates student responses automatically, and persists results to the database.

D. Knowledge Level Classification

Based on assessed quiz scores, each student is classified per module into one of three levels: Basic, Intermediate, or Advanced. This classification forms the basis for all downstream personalization in resource recommendation and timetable generation.

E. Learning Resource Recommendation

Based on the identified knowledge level per module, the system recommends appropriate learning resources including PDF study materials and curated YouTube video tutorials, accessible directly from the student's timetable view.

F. Personalized Timetable Generation

The system generates a personalized study timetable for each student, with daily study hour allocations differentiated by knowledge level: more time for Basic, moderate for Intermediate, and less for Advanced modules. Examination proximity is also factored into the scheduling logic.

G. Real-Time Chat Communication

Students can send and receive messages in a shared real-time chat room, enabling peer collaboration, study group formation, and collective doubt resolution across the student community.

V. NON-FUNCTIONAL REQUIREMENTS

Performance: The system delivers fast response times for authentication, quiz submission, assessment evaluation, and resource loading operations.

Security: Secure user authentication is enforced, and all user data is stored in a protected database with controlled access.

Usability: The interface is designed to be simple and intuitive, minimizing the learning curve for students unfamiliar with digital learning platforms.

Scalability: The architecture supports concurrent access by multiple students and handles simultaneous chat interactions without degradation of service.

Reliability: Assessment evaluation and knowledge level classification operate with high accuracy and consistency across repeated interactions.

Availability: The system is accessible at any time through a standard web browser without requiring additional software installation.

Maintainability: Administrators can readily update subjects, modules, questions, and study materials through the admin portal without requiring code changes.

VI. RESULTS

EduPath was successfully implemented and evaluated across its core functional modules. The platform enables students to take module-level knowledge assessment quizzes spanning multiple subjects. Upon quiz completion, the system correctly classifies each module into the appropriate level—Basic, Intermediate, or Advanced—based on the student's performance profile. The generated study timetable accurately reflects these classifications, with daily study hour allocations proportional to identified knowledge gaps and examination proximity.

Level-appropriate learning resources—PDF notes and YouTube tutorials—are correctly surfaced in the timetable view for each module. The real-time chat room operates as intended, enabling students to exchange messages and engage in peer collaboration. The administrative interface supports full content management, allowing administrators to manage subjects, modules, quiz questions, and study materials through a clean and functional portal.

The system was tested with sample student data across subjects including AAD, CG, Python, and ML. Classification accuracy, timetable generation correctness, and resource retrieval were verified through systematic functional testing.

VII. CONCLUSION

EduPath demonstrates that a lightweight, web-based personalized study guidance system can meaningfully improve the way engineering students approach examination preparation. By automatically assessing knowledge levels through module-level quizzes, generating adaptive study timetables, and recommending level-appropriate learning resources, the platform converts a generic study experience into one that is tailored to each student's individual strengths and weaknesses. The inclusion of a real-time peer chat facility further supports collaborative learning within the student community.

The system's technical implementation—using React.js, Flask, and SQLite—confirms that effective academic personalization can be achieved with accessible, open-source technologies. EduPath aligns with broader goals of equitable and quality education by reducing the dependency on external tutoring and enabling students to direct their own learning with data-driven guidance.

Future enhancements will focus on integrating an AI-powered chatbot for instant doubt clarification, implementing dynamic study plan updates based on ongoing performance tracking, adding advanced analytics for identifying weak topic patterns, extending the platform to mobile applications, and incorporating a broader library of interactive educational resources.

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